A3 — Feedback Data and Lumped Modeling

PHYS 39 · Module 6 · Kyle Chen and Muhammad Hasan · October 19, 2026

1. One-lump model and P-control predictions

The First Law gives storage = TEC heat input – heat loss to the room. With C = mc_p, constant thermal properties, and signed applied command u:

$$C\frac{dT}{dt} = P_u u - H(T - T_{amb})$$

C is in J/K, Pu in W/PWM, and H in W/K; each term has units W. At constant command, the steady balance gives Tss = Tamb + (Pu/H)u. Comparing with measured susceptibility identifies $\chi = Pu/H$. Subtracting steady state gives $C\, d(T - Tss)/dt = -H(T - Tss)$, so $\tau = C/H$.

$$\frac{dT}{dt} = \frac{T_{amb} + \chi u - T}{\tau}, \qquad \tau = 91.69 \text{ s}$$

Measured $\chi h = 0.5311$ and $\chi c = 0.1878$ °C/PWM. Heating and cooling coefficients differ because resistive heating aids heating and opposes cooling. The 63% heating crossing estimates τ to about one 0.57 s sampling interval; ambient drift and incomplete settling add unquantified uncertainty.

Substitute P control u = Kp(Tset – T). At steady state, $(1 + \chi Kp)Tss = Tamb + \chi KpTset$. Subtract the steady balance to obtain the transient:

$$e_{ss} = \frac{T_{set} - T_{amb}}{1 + \chi K_p}, \quad L = \chi K_p, \quad \tau_{cl} = \frac{\tau}{1 + L}$$

L is dimensionless. Increasing Kp reduces both droop and response time by 1 + L. P requires nonzero error to replace heat lost to the room. At Kp = 10 and target 30 °C, the saved baseline predicts droop 0.916 °C and $\tau cl = 14.53$ s.

Measured check: for +5 PWM and Tamb = 24.220 °C, predicted Tss = 26.875 °C versus a measured final-minute mean of 26.366 °C. Warming and approximate scale agree; the 0.510 °C discrepancy exposes model/calibration limits. The same trace supplied τ, so timing agreement is not independent evidence.

Module 5 interpretation: insulation lowers H and droop; doubling C slows the transient without changing the steady balance. Kp = 1 gives L = 0.5311, not much smaller than one; L = 1 at Kp = 1.883 PWM/°C. Module 5 gain 20 approximately settled at 29.386 °C (droop 0.614 °C), while higher gains showed increasingly ambiguous or recurring waves. Different ambient conditions limit quantitative cross-day comparisons.

2. Experimental Goldilocks plots and metrics

Six recorded runs use the same apparatus, 28 → 30 → 28 °C setpoints, PWM limit ±255, anti-windup, and ≈0.57 s sampling. Each has about 10 s baseline and 40 s per step. The integral was reset before each run. Runs 3 and 6 are the October 19 Kp = 1000 reruns; the others are October 14 recordings.

Protocol limitation: starting temperatures differ, and selected PI gains are 16/0.1, 16/0.5, and 1000/2 (Kp/Ki). Thus Run 6 does not isolate increasing Ki at fixed Kp. We preserve and disclose the actual recordings rather than presenting them as a matched Ki sweep.

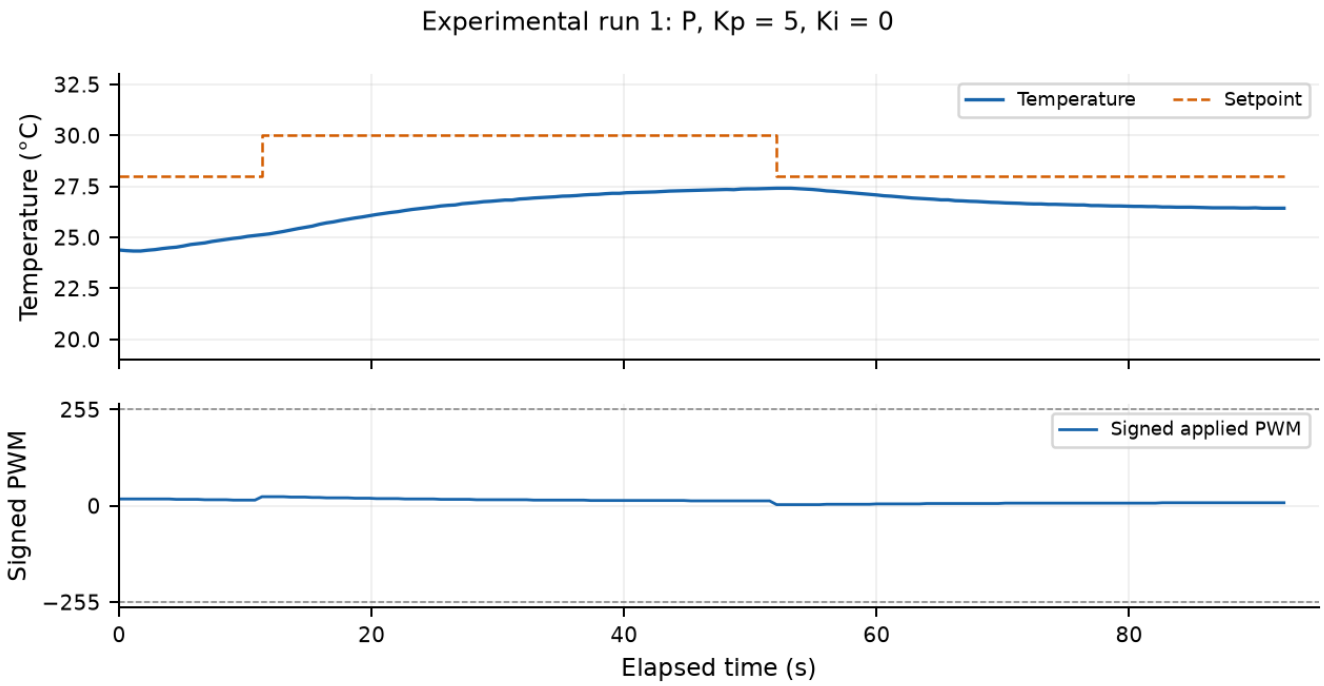


Figure 1. Experimental P: Kp = 5 PWM/°C; Ki = 0 PWM/(°C·s). Initial T = 24.38 °C; 0/164 applied commands at ±255. Only actual applied PWM is shown; controller requests are not PWM output.

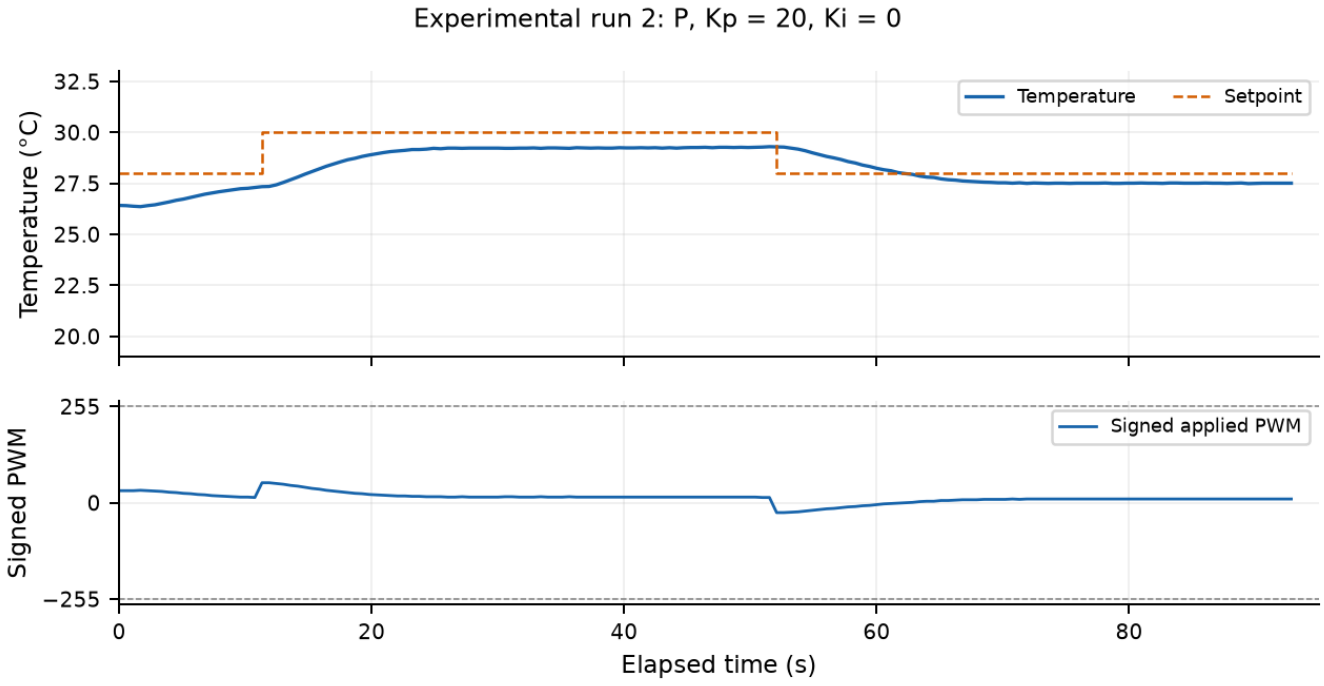


Figure 2. Experimental P: Kp = 20 PWM/°C; Ki = 0 PWM/(°C·s). Initial T = 26.42 °C; 0/165 applied commands at ±255. Only actual applied PWM is shown; controller requests are not PWM output.

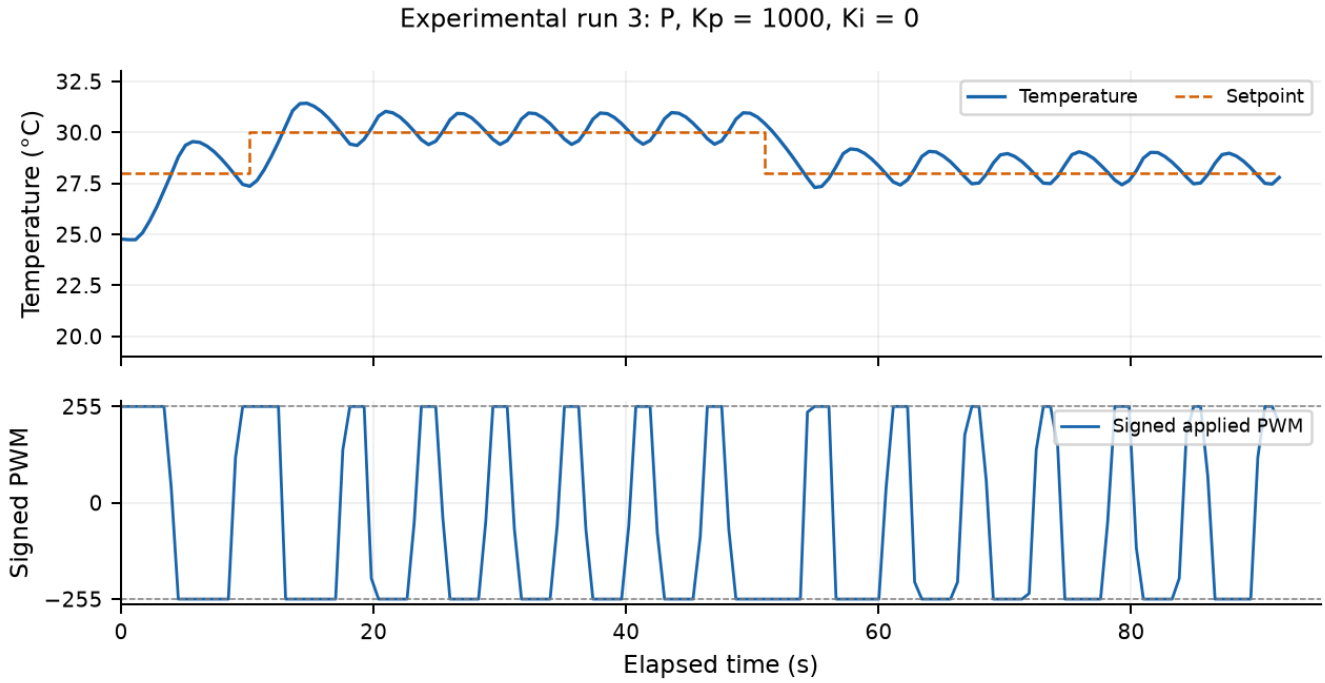


Figure 3. Experimental P: Kp = 1000 PWM/°C; Ki = 0 PWM/(°C·s). Initial T = 24.77 °C; 133/163 applied commands at ±255. Only actual applied PWM is shown; controller requests are not PWM output.

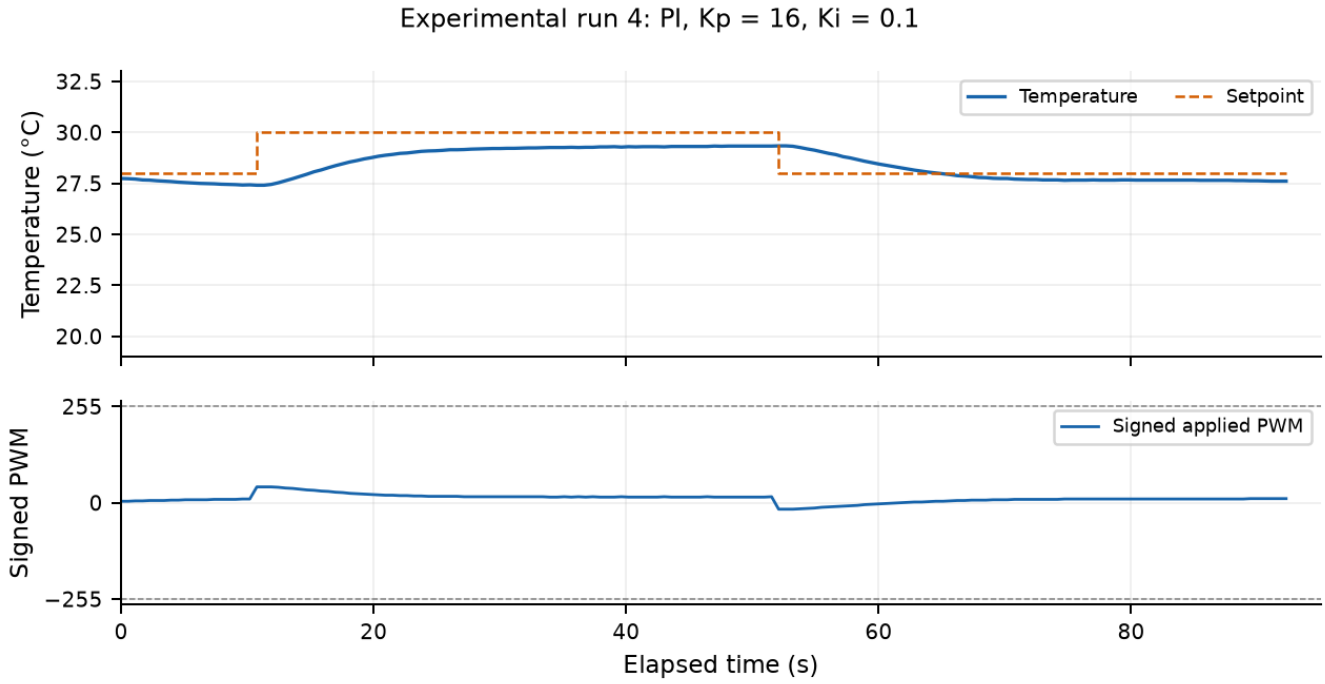


Figure 4. Experimental PI: Kp = 16 PWM/°C; Ki = 0.1 PWM/(°C·s). Initial T = 27.74 °C; 0/164 applied commands at ±255. Only actual applied PWM is shown; controller requests are not PWM output.

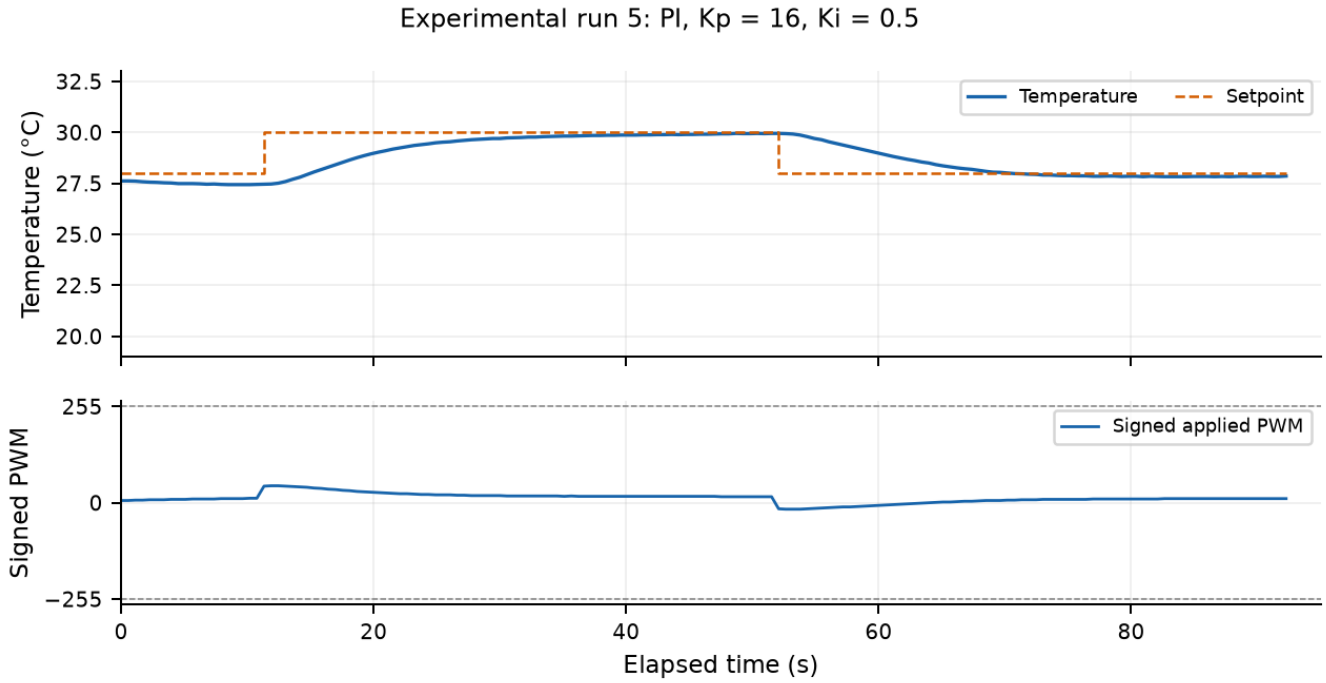


Figure 5. Experimental PI: Kp = 16 PWM/°C; Ki = 0.5 PWM/(°C·s). Initial T = 27.62 °C; 0/163 applied commands at ±255. Only actual applied PWM is shown; controller requests are not PWM output.

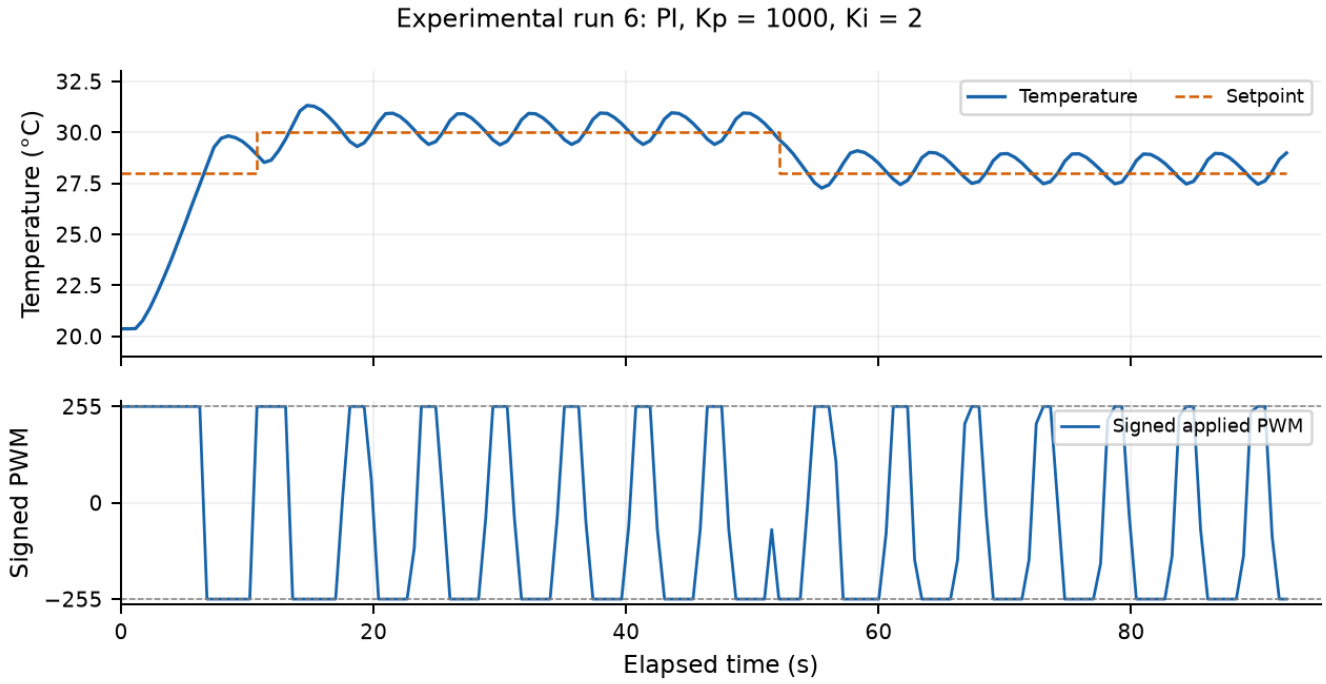


Figure 6. Experimental PI: Kp = 1000 PWM/°C; Ki = 2 PWM/(°C·s). Initial T = 20.37 °C; 132/164 applied commands at ±255. Only actual applied PWM is shown; controller requests are not PWM output.

Largest-gain P and PI: both step directions

Rise time is the first 10%–90% crossing from the last pre-step measured temperature toward the new target, interpolated between samples. Overshoot is excursion beyond the target divided by the commanded 2 °C step. Settling requires remaining within ±0.04 °C through the end of the step, with at least 10 s observed in-band.

Mode Kp/Ki	Step	Rise (s)	Overshoot (%)	Period (s)	End error (°C)
P 1000/0	Up	1.79	71.5	5.67	-0.240
P 1000/0	Down	2.82	35.0	6.23	-0.247
PI 1000/2	Up	0.54	66.0	5.66	-0.317
PI 1000/2	Down	2.03	36.5	5.67	-0.221

For all four high-gain steps, settling time and steady-state error are **not established**. End error is target minus the final 5 s mean, not steady-state error. The 0.54 s PI upward rise spans approximately one sample interval and is poorly resolved. Period is the median peak-to-peak spacing, requiring at least three peaks confirmed by reversals ≥0.20 °C; sampling limits timing resolution to roughly 0.57 s.

Both high-gain recordings have recurring, bounded oscillations over the observed window. This is consistent with a nonlinear limit cycle, but does not prove ideal marginal stability or long-term boundedness. P run 3 reaches 31.43 °C / 27.30 °C on the up/down steps; PI run 6 reaches 31.32 °C / 27.27 °C. Applied PWM saturates on 133/163 and 132/164 samples, respectively.

3. Two-lump simulation and experiment comparison

One shared physical parameter set for all six simulations: $\chi c = 0.1878$ and $\chi h = 0.5311$ °C/PWM; $\tau = 91.69$ s; Tamb = 24.220 °C; C = 100.549 J/K; H = 1.0966 W/K; Pu,c = 0.20595 W/PWM; r = 2.8280; G = 2 W/K; CT/C = 0.35; TEC voltage 12 V (assumed ON-state value), resistance 1.50 Ω; both masses initially 28 °C; targets 28 → 30 → 28 °C at 11 and 52 s; PWM ±255; q(0) = 0; anti-windup on; $\Delta t = 0.01$ s.

G and CT/C are unresolved internal estimates, retained at the assignment starting values. The high-P model reproduces overshoot and decay qualitatively, but cannot reproduce the measured persistent wave train. Exact fitting is not claimed. All simulations keep the same initial state as required; their 28 °C start differs from the recorded starts, and the common switching times differ slightly from individual host-timed experiments.

$$C_T\dot{T} = P_u u - G(T - T_m), \qquad C_m\dot{T}_m = G(T - T_m) - H(T_m - T_{amb})$$

The controller measures Tm. With perturbations θT and θm about equilibrium and unsaturated P feedback, GθT = Cm dθm/dt + (G + H)θm. Differentiate this expression and substitute the TEC balance to eliminate θT:

$$C_T C_m\ddot{\theta}_m + S\dot{\theta}_m + G(H + P_u K_p)\theta_m = 0, \quad S = C_T(G + H) + C_m G$$

$$\zeta_{2L,P} = \frac{S}{2\sqrt{C_T C_m G(H + P_u K_p)}}, \quad K_{p,crit} = \frac{S^2/(4C_T C_m G) - H}{P_u}$$

For this shared plant, heating-direction Kp,crit = **3.478 PWM/°C**. The three gains 5, 20, and 1000 are all above it, with progressively smaller damping. This shared model therefore does not reproduce a monotonic low-gain-to-critical-to-underdamped progression for these selected experimental gains.

$$C_T C_m s^3 + Ss^2 + G(H + P_u K_p)s + GP_u K_i = 0$$

For positive coefficients, Routh–Hurwitz requires S·G(H + PuKp) > CT·Cm·G·Pu·Ki. This yields:

$$K_{i,crit} = \frac{S(H + P_u K_p)}{C_T C_m P_u}, \qquad \rho_I = K_i/K_{i,crit}$$

The unsaturated local PI model is stable for $0 < \rho I < 1$, marginal at $\rho I = 1$, and unstable for $\rho I > 1$. These heating-equilibrium classifications do not describe actuator clipping or the complete heating/cooling switching transient.

Six simulations using the same plant and protocol

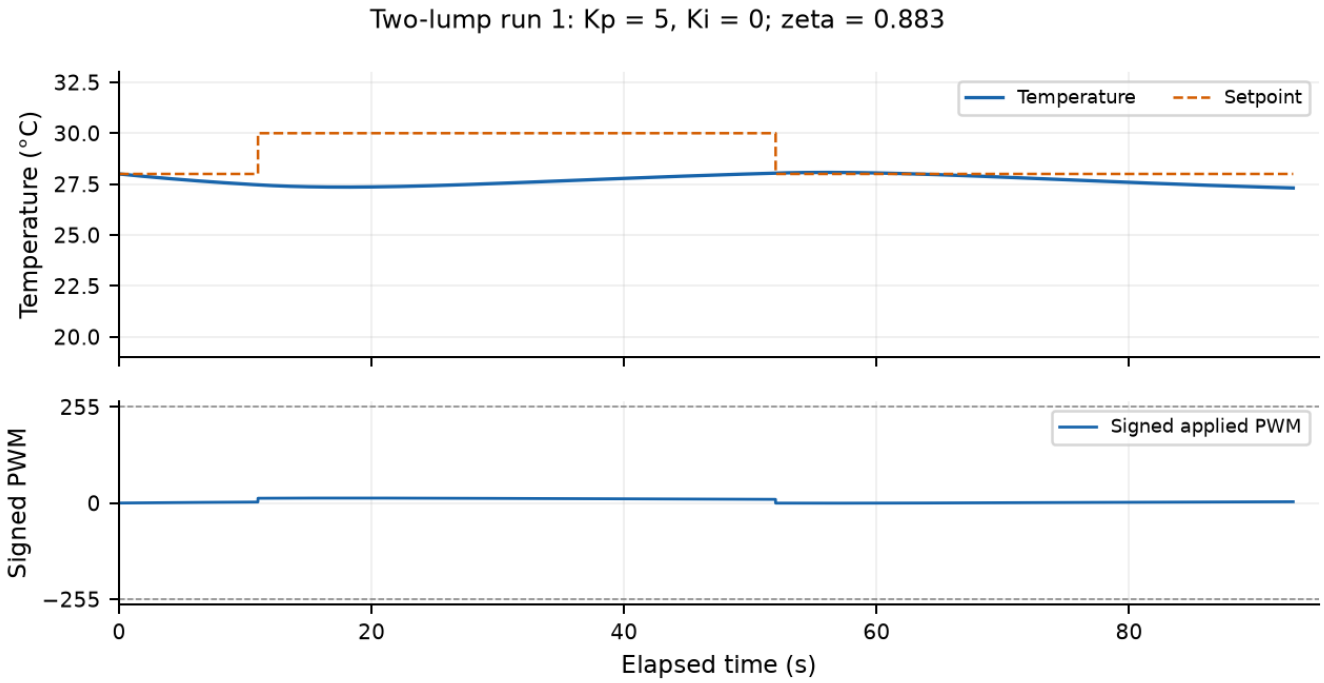


Figure 7. Two-lump P: Kp = 5, Ki = 0; $\zeta_{2L,P}$ = 0.883. Local classification: stable underdamped. Applied PWM is clipped at ±255; nonlinear portions are not classified by the linear criterion alone.

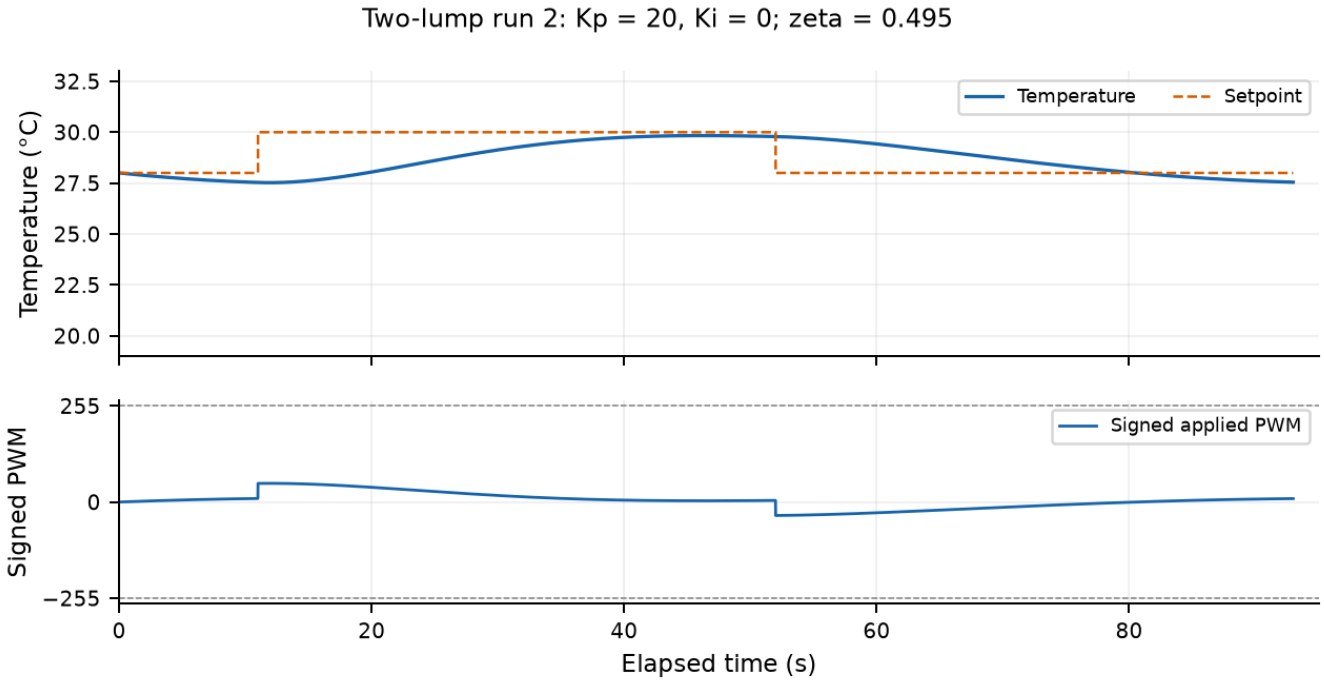


Figure 8. Two-lump P: Kp = 20, Ki = 0; $\zeta_{2L,P}$ = 0.495. Local classification: stable underdamped. Applied PWM is clipped at ±255; nonlinear portions are not classified by the linear criterion alone.

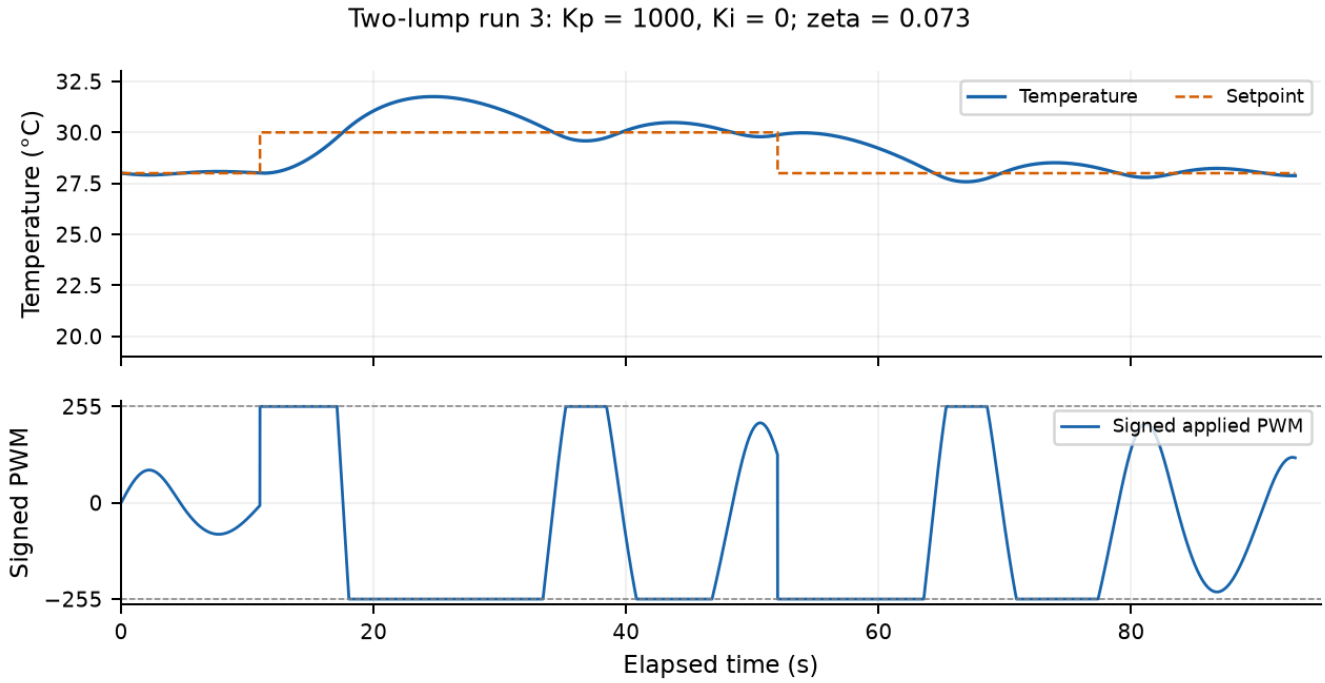


Figure 9. Two-lump P: Kp = 1000, Ki = 0; $\zeta_{2L,P}$ = 0.073. Local classification: stable underdamped. Applied PWM is clipped at ±255; nonlinear portions are not classified by the linear criterion alone.

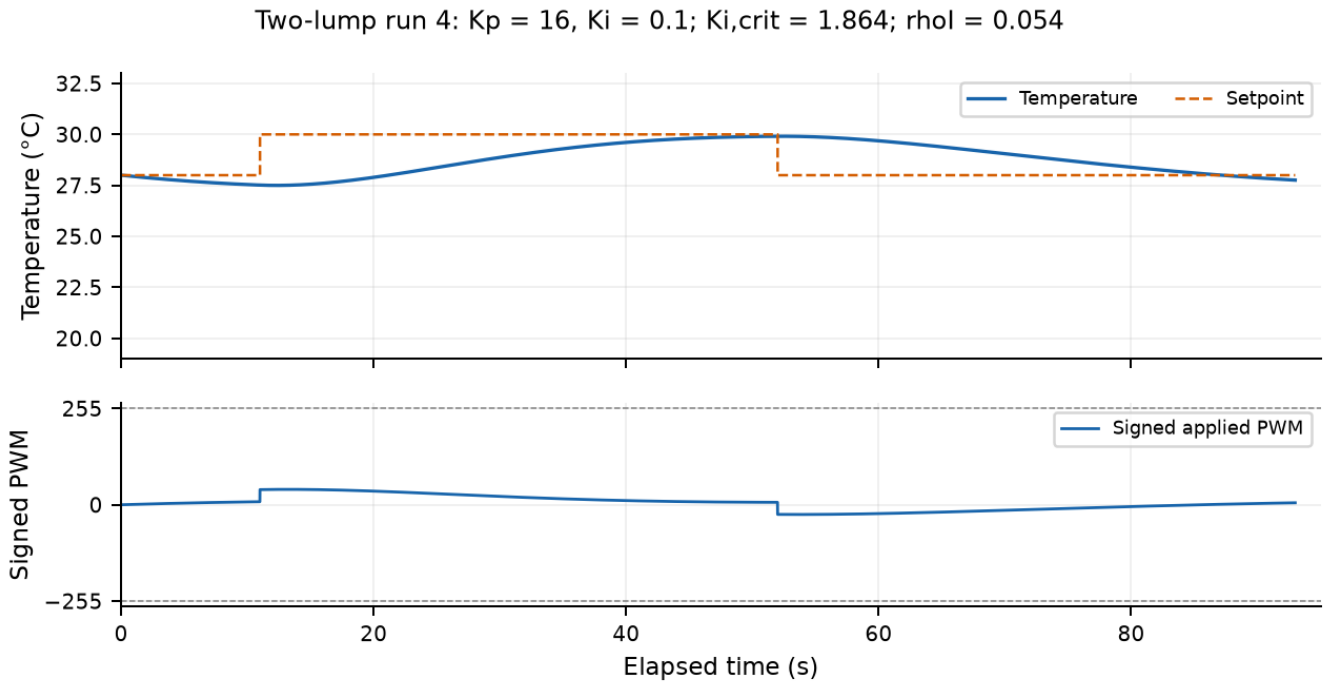


Figure 10. Two-lump PI: Kp = 16, Ki = 0.1; Ki,crit = 1.864; ρI = 0.054. Local classification: stable underdamped. Applied PWM is clipped at ±255; nonlinear portions are not classified by the linear criterion alone.

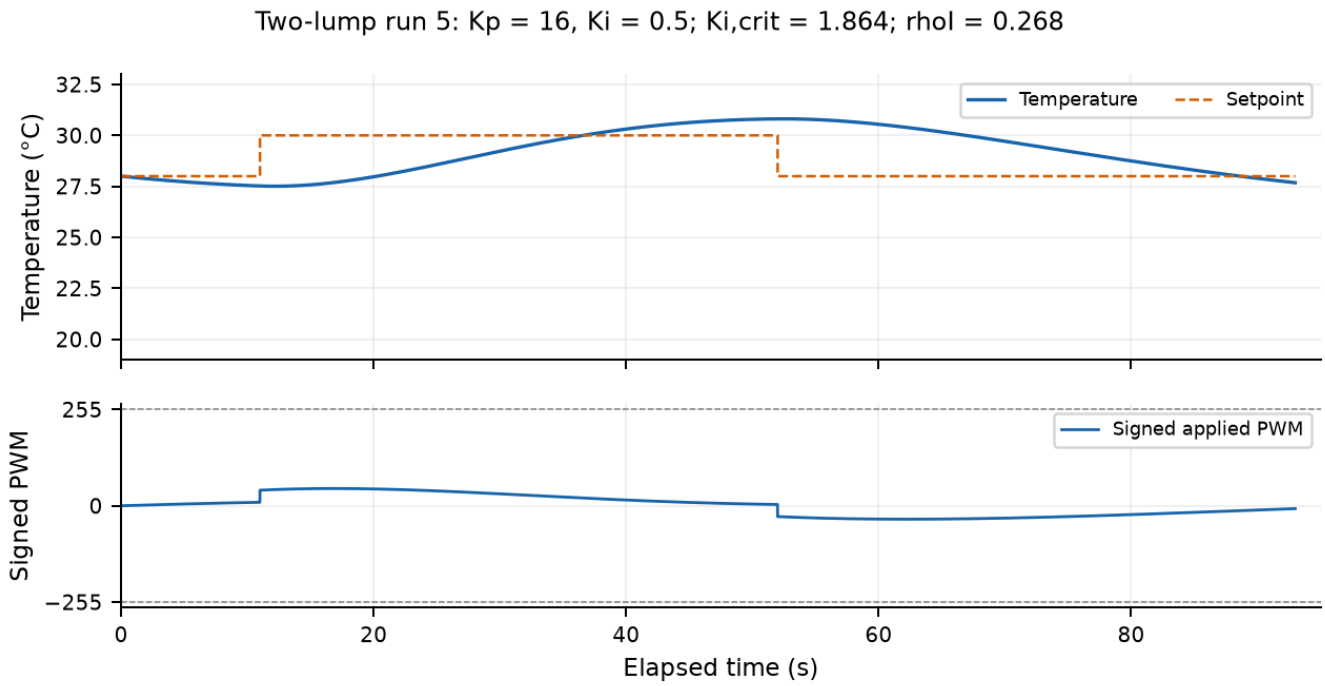


Figure 11. Two-lump PI: Kp = 16, Ki = 0.5; Ki,crit = 1.864; ρI = 0.268. Local classification: stable underdamped. Applied PWM is clipped at ±255; nonlinear portions are not classified by the linear criterion alone.

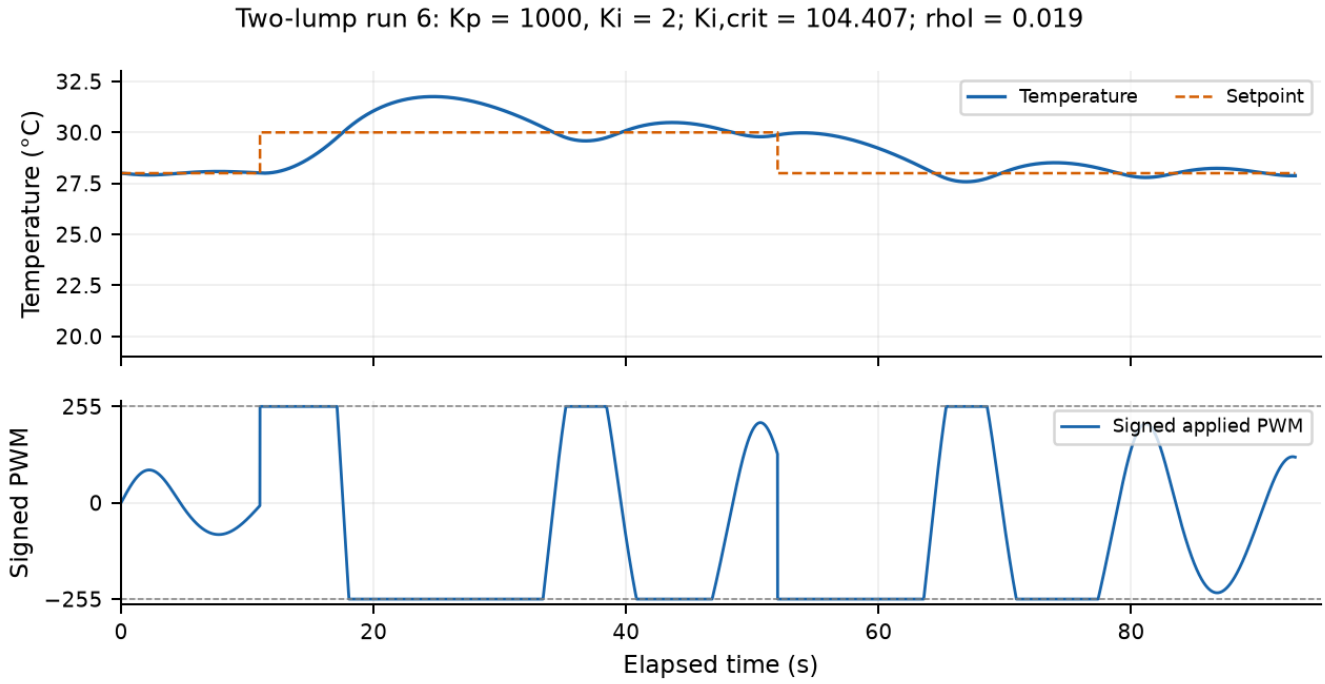


Figure 12. Two-lump PI: Kp = 1000, Ki = 2; Ki,crit = 104.407; ρI = 0.019. Local classification: stable underdamped. Applied PWM is clipped at ±255; nonlinear portions are not classified by the linear criterion alone.

Guided comparison answers

Run	Kp/Ki	Stability measure	Local model class
1	5/0	0.883	stable underdamped
2	20/0	0.495	stable underdamped
3	1000/0	0.073	stable underdamped
4	16/0.1	ρI = 0.054	stable underdamped
5	16/0.5	ρI = 0.268	stable underdamped
6	1000/2	ρI = 0.019	stable underdamped

1. Best correspondence: low/intermediate P simulations reproduce slow approach and remaining droop qualitatively; intermediate PI (Run 5) reproduces reduced target error. None reproduces the complete high-gain speed, persistent oscillation, and settling behavior. Initial conditions prevent a fair numerical ranking by rise time alone.

2. P organization: ζ decreases from 0.883 to 0.495 to 0.073; all gains exceed Kp,crit. Linear P poles remain negative, so ringing decays. The simulation can clip at high gain, requiring the nonlinear time trace as well as the pole calculation.

3. PI organization: Runs 4–5 share Kp = 16 and increase ρI from 0.054 to 0.268, making integral correction more aggressive. Run 6 changes Kp to 1000, raising Ki,crit to 104.407 and reducing ρI to 0.019 despite its higher Ki. The retained sequence is not an increasing-ρI study. All three local linear models are stable underdamped; none is marginal or unstable.

High-gain case	Measured rise (s)	Model rise (s)	Measured OS (%)	Model OS (%)
P Up	1.79	4.04	71.5	87.8
P Down	2.82	6.22	35.0	21.0
PI Up	0.54	4.04	66.0	87.8
PI Down	2.03	6.22	36.5	21.0

4. Agreement and disagreement: both model and experiment show overshoot and PWM clipping at high gain. The measured high-gain period is about 5.7–6.2 s, whereas the shared model has slower, decaying oscillations. Persistent measured ringing contradicts the ideal unsaturated, continuous-time prediction of decay if that model is used to explain the full experiment. It does not prove the apparatus is linearly marginal: saturation, sampled control, additional thermal states, and changing parameters are omitted effects.

The high-gain simulations are model tests rather than fitted predictions. Halving Δt to 0.005 s changes temperature by at most 0.0201 °C across the six runs, much smaller than their main experimental discrepancies.

Controlled parameter checks: one-lump PI comparisons show higher Ki and C lower damping, while higher Kp raises it. The assigned two-lump P cases give ζ ≈1.31, 1.00, 0.81 at Kp = 60, 105, 160. At Kp = 105, G = 35, 50, 70 W/K gives ζ ≈0.84, 1.00, 1.18. An approximately equal capacitance split gives the weakest damping among the tested splits. These practice figures remain in Part_II/RESULTS.md, separate from the twelve Goldilocks figures.

4. PI steady state, assumptions, and windup

$$u = K_p e + K_i q, \quad \dot{q} = e, \quad e = T_{set} - T$$

At a stable, unsaturated PI equilibrium, q is constant, so dq/dt = e = 0. Therefore **Tss = Tset = 55 °C**. The P term becomes zero, while the stored I term supplies the heat-loss-balancing command:

$$u_{ss} = u_I = \frac{H(55 - T_{amb})}{P_{u,h}} = \frac{55 - 24.220}{0.5311} = 57.95 \text{ PWM}$$

This is an ideal calculation, not a 55 °C hardware run. It requires stable feedback, enough heating power within ±255, no active shutdown, continued integral action with Ki > 0, and accurate, timely measurement. Sensor bias would make indicated rather than actual temperature equal 55 °C.

Windup: while applied PWM is capped, persistent error can keep increasing q. The excessive stored command then continues heating after the target is crossed, causing overshoot and delayed recovery. Conditional anti-windup holds q when the proposed integral update pushes the request farther past the limit, and permits updates that bring it back. The clamp and shutdown remain separate protections.

Reset thought experiment: at equilibrium P = 0 and I provides the balancing heat. Clearing q removes that command; temperature initially falls, then P responds and I rebuilds. Integral reset does not preserve the former equilibrium output.

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5. Model limits and final gain choice

$$T(t) - T_{ss} = [T(0) - T_{ss}]\exp[-(1 + \chi K_p)t/\tau]$$

In the ideal continuous-time one-lump P model, the exponential is positive and decays: temperature approaches its equilibrium monotonically and cannot ring. This statement assumes instantaneous feedback; it is not a claim that every physical P controller is non-oscillatory. The observed Kp = 1000 P oscillation is direct evidence that this simplest model is inadequate.

The two-lump balances add a second stored-energy state. Heat first changes the TEC-side temperature, then passes through G to the measured mass. Tm can continue rising after the command falls because the TEC-side mass remains hotter. This lag permits P overshoot and ringing; the twelve figures show what the model captures and where it fails.

Selected P: Kp = 20 PWM/°C is a provisional measured compromise: it reduces the large error of Kp = 5 and avoids the large oscillations and saturation of Kp = 1000. The selected shared model gives ζ = 0.495 and Kp,crit = 3.478, predicting an underdamped response rather than critical damping. Selection is based on the measured tradeoff, not a claim that this gain is theoretically optimal.

Selected PI: Kp = 16 PWM/°C and Ki = 0.5 PWM/(°C·s), with ρI = 0.268, combines relatively small final target error with limited overshoot in Run 5. Ki = 0.1 corrects more slowly; Kp = 1000, Ki = 2 produces persistent large excursions and clipping. Since the high-PI rerun changes both gains, it cannot establish the isolated effect of Ki. Neither selected gain pair has a verified long-term settling result.

Next physical effect: include the actual ≈0.57 s sampled controller and measurement/command delay. The present continuous-feedback simulation updates much faster. Further thermal states, sensor placement, and temperature-dependent TEC coefficients may also matter; these data do not identify a unique cause of the persistent oscillation.

Data, code, and reproduction

Build command from the repository root (phys39a environment):
/opt/anaconda3/envs/phys39a/bin/python code/module6/A3/build_a3.py --save-csv

Analysis and report builder · Experimental metrics · Shared settings and stability calculations · Controlled comparison results

Experimental Run 1 raw recording · Simulation Run 1 data

Experimental Run 2 raw recording · Simulation Run 2 data

Experimental Run 3 raw recording · Simulation Run 3 data

Experimental Run 4 raw recording · Simulation Run 4 data

Experimental Run 5 raw recording · Simulation Run 5 data

Experimental Run 6 raw recording · Simulation Run 6 data

GitHub modeling/data checkpoint: ff2f524 — Part 7 data with saturation reruns. This report is a later local build; the checkpoint is not claimed to include the finished PDF.

Assignment: A3 Questions 1–5. Simulation protocol: Module 6 Part II.